

Analysis — AdamW vs Muon on SST-2

We compare the tuned (“best”) AdamW and Muon runs from Phase 2 (5,000 steps), with the best checkpoint selected by validation accuracy. Validation accuracy is the primary comparison criterion because the official SST-2 validation split is independent of the training data and was also used as the Optuna objective.

Run	Best step	Val. acc.	Test acc.	Grad. norm	Rel. update	Mom. cosine	Probe cosine	Sharpness @ $\epsilon=.05$
AdamW	1600	0.9106	0.9355	3.81	1.31e-4	0.002	0.932	-0.0215
Muon	3800	0.9025	0.9500	2.74	2.49e-4	0.028	0.967	-0.0080

Why these metrics?

- Gradient norm - shows whether the optimizers operate with different gradient magnitudes.
- Relative update - shows differences in how each optimizer transforms gradients into updates.
- Momentum–gradient cosine - measures the consistency with recent optimization history.
- Probe-gradient cosine (on a fixed batch) - indicates the stability of the optimization direction.
- Sharpness (under filter-normalized perturbations), with lower values indicating a flatter response.

Results

AdamW wins on the main metric, achieving a validation accuracy of 0.9106 compared to 0.9025 for Muon. Muon nevertheless produces different optimization dynamics: despite a smaller gradient norm, it makes nearly twice the relative update and has higher momentum and probe-gradient alignment.

AdamW also has the lower sharpness value (−0.0215 vs. −0.0080) and lower post-checkpoint validation-loss volatility (0.0408 vs. 0.0457), indicating a more stable local solution under our measurements. These geometric differences indicate better generalization.

The held-out train slice instead favors Muon (0.9500 vs. 0.9355). However, it is not an independent test distribution: it was taken from the training split and contains the same short-phrase-heavy distribution. Therefore, validation accuracy is a better measurement for comparing optimizers.

Muon is also sensitive to the learning rate: its matched default LR of 0.02 produced only 0.8131 validation accuracy, whereas Phase 1 selects 1.03e-3 and much higher results.

Reliability

The study uses a single seed, a single architecture, and a single benchmark. A stronger study should use multiple seeds and an independent test set. Moreover, harder/larger tasks and models should be tested, as this fine-tuning training achieved immediate results. Longer training and harder tasks might produce different results. Therefore, a reliable study should explore and test variations across these aspects.

Conclusion: AdamW gives the best validation performance in this experiment, while tuned Muon achieves higher in-distribution held-out accuracy and exhibits distinctly different optimization dynamics. The results show meaningful differences among optimizers, but are insufficient to establish a clear winner.